

Bitla Umesh Kumar

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Professional Summary

CSE student specializing in AI/ML and full-stack development. Proven track record building production-grade RAG systems, AI-native hackathon projects, and scalable web applications across multiple internships. Founder of SVIT's LEGION Club.

Technical Skills

Languages: Python, C, HTML, CSS

Dev Tools: Supabase, Git, Docker, Upstash Redis

Frameworks & APIs: React, Next.js, FastAPI, Streamlit, ChromaDB, PyMuPDF, Groq API, Cohere API, OpenRouter, pgvector

Experience

Alonzo AI – AI & Full-Stack Engineering Intern (SPARK Program) May 2026 – June 2026

- Merit-selected (5 students/college) for an in-person industry internship at Gachibowli, Hyderabad; building production-grade AI and full-stack projects with Alonzo AI's engineering team.
- Contributing to open-source repos via Agentic Driven Development practices and delivering a capstone project at program conclusion.

Squarcell Resource India (QSkill) – AI/ML Intern Apr 2026 – May 2026

- Selected for an intensive virtual AI/ML program focusing on hands-on skill development and practical model deployment.

YugaYatra Retail OPC Pvt. Ltd. – Software Engineer Intern Mar 2026 – May 2026

- Developed scalable web applications utilizing Cursor AI, Supabase, and integrated AI technologies for live freelance projects.
- Contributed to e-commerce deployments and assisted in seller portal management, ensuring stable production rollouts.

LEGION Club (SVIT) – Founder & Non-Technical Lead Feb 2026 – Present

- Founded and structured the CSE Department club into Technical and Non-Technical divisions to serve the student body.
- Directed the Non-Technical division, orchestrating multiple department-wide events and managing all logistical operations.

Projects

Natya Samhita – Sanskrit RAG Platform for Natya Shastra github.com/BitlaUmesh/Natya-Samhita

- Built a full-stack RAG platform (Next.js / FastAPI) for semantic querying of Sanskrit treatises, using pgvector HNSW embeddings, Cohere two-stage reranking, and DeepSeek V3.1 for grounded generation.
- Integrated Upstash Redis caching (24h TTL) to cut repeated query latency to ~15ms and reduce external API costs.

Neural-Sync – 4th Place, HackForge 48-Hour Hackathon github.com/BitlaUmesh/Neural_Sync

- Built a Python framework synchronizing neural network outputs with high-frequency data streams; optimized pipelines to reduce real-time inference latency.
- Secured 4th Place overall and a Software Engineering Internship offer by developing and presenting this framework.

Vision-RAG – Multimodal Document Intelligence System github.com/BitlaUmesh/Vision-RAG

- Engineered a full-stack mRAG framework integrating computer vision and LLMs to query visually complex PDFs via FastAPI, PyMuPDF, and Groq API — eliminating OCR dependency.
- Implemented hybrid similarity search with ChromaDB and local sentence-transformers to suppress hallucinations.

Education

Swami Vivekananda Institute of Technology, B.Tech in Computer Science Sept 2023 – May 2027

- **GPA:** 7.2/10.0
- **Relevant Coursework:** Data Structures, Computer Networking, DevOps, DBMS